Thomas Chisica Londoño

Applied Mathematics & Computer Science | Bogotá, Colombia

thomas.chisica@urosario.edu.co | Thom-320

Research Profile

Applied mathematics student working on AI evaluation and measurement validity: replicating capability measurements, LLM-judge stability, and RL policies tested against calibrated baselines, with held-out evaluation and explicit limits on what results show.

Education

Universidad del Rosario, Bogotá 2024–present

BSc in Applied Mathematics and Computer Science. Expected graduation: June 2028.
GPA: 4.11/5.00 | Full-tuition merit scholarship.

Relevant coursework: Machine Learning, Probability & Statistics, Real Analysis, Linear Algebra, Graph Theory, Differential Equations, Optimisation and Algorithms & Data Structures.

Research Experience

AI evaluation and measurement validity 2026–present

Independent projects

- Replicated METR's Time Horizon 1.1 from its public code and data (all 20 models in the public runs file match; 128.7-day doubling). With only measured human anchors (162 of 228 tasks), the top model's 50% horizon is not identified, while the doubling time barely moves (exploratory pilot). Code and memo
- On 120 Petri calibration transcripts, three LLM judges reached ICC 0.74–0.88, yet on misuse cooperation (ICC 0.81) the target order held under all judges in only 34% of bootstrap draws; found silent scoring defects in Petri. Pre-registration for the main study drafted. Pilot

Simulation and reinforcement learning for supply-chain resilience 2025–present

Undergraduate research with Prof. Alexander Garrido, Universidad del Rosario

- Reconstructed Garrido-Ríos's (2017) thirteen-operation supply-chain model in SimPy with Gymnasium environments for PPO, tests and source-model checks.
- Benchmarked 10 PPO checkpoints on 60 held-out streams against a calibration-selected static policy under the same decision contract; PPO trailed it (ahead in 2 of 10 checkpoint means), so the apparent RL advantage was withdrawn.

SCRES: simulation and evaluation code

Collective behaviour and human coordination 2025–present

Independent reanalysis; software for Prof. Edgar Andrade-Lotero's lab, Universidad del Rosario

- Reanalysed Andrade-Lotero and Goldstone's (2021) division-of-labour experiment: modelled the 8×8 board as a graph, rebuilt past-only features for 45 dyads and checked temporal integrity with six tests.
- Certified the graph's exact minimum conductance (22/210, attained only by the four half-board cuts) and implemented basis-invariant Fiedler-subspace projectors. Spectral and coordinate energies were nearly redundant; predictive advantage remains unestablished.
- Simulated identifiability: in table form the rival models were not identifiable with this design; parametric rewrites helped, but not for every parameter.
- For the lab's PsyNet experiments, wrote docs and smoke tests and built tutorials, task maps and a score screen for the foraging_trucks demo (merged October 2026).

Spectral/SODCL: reanalysis and temporal audit

Recurrent neural networks: connectivity and motor learning July–October 2026

Neuromatch Academy Computational Neuroscience (68 h): team project and independent follow-up

- Ran Python/NumPy experiments on 200-unit FORCE-trained recurrent networks across four connectivity densities and 16 paired seeds, with learning disabled during held-out evaluation.

– Added an equal-plasticity control: with the same trainable-edge budget at every density, the density advantage disappears (16/16 seeds). An earlier 8-seed reversal did not hold up with more seeds.

Motor-RNN experiments and equal-plasticity control

Animal social networks: yellow-bellied marmots 2026–present

Research with Dr. Adriana Maldonado-Chaparro, Universidad del Rosario

– Built observation-aware SQLite dyadic tables (interactions, sightings, residence, relatedness); exploratory count/mixed-effects models and matched samples to examine sampling effects.
– Validated a legacy NetLogo/R individual-based model with Python and native-runtime checks; found homophily-rule defects and documented reproducibility limits.

Technical Skills

Programming & ML: Python, PyTorch, Gymnasium; C, R, SQL; JavaScript (React, Node.js).
Scientific computing: NumPy/SciPy, SimPy, NetworkX, statsmodels; NetLogo model audits.
LLM systems & software: RAG, embeddings, FAISS, FastAPI; Git, Linux, pytest/CI, SQLite, PostgreSQL, Polars, LaTeX.
Languages: Spanish (native); English C1 (IELTS General Training 7.0, 2023).

Teaching, Awards & Funding

Calculus I Teaching Assistant, Universidad del Rosario July 2024–July 2025

Led problem-solving sessions and supported first-year students.

Andrés Bello National Distinction 2024

Colombian Ministry of Education

Ranked 33rd nationally in the official distinction ranking for the 2023 Saber 11 examination (Colombia's national school-leaving examination); one of 50 national recipients. Highest-ranked recipient from Tunja, which ranked first among certified education authorities in the reported 2023 results.

Full-tuition merit scholarship, Universidad del Rosario 2024–present

Additional Experience

Data Analyst Trainee, Blend360 Colombia (offer declined) 2026

Passed every selection stage and was offered a contract; declined it to focus on my degree and research.

Lead in Emerging Technologies (AI), Witty June–December 2025

Witty and Witty Wallet, the digital platform of Progresión Sociedad Comisionista de Bolsa S.A.

Led emerging-technology work (AI, analytics, data engineering). Built a retrieval-augmented support prototype (FastAPI, embeddings, optional FAISS, LLM rewriting) with heuristic routing that either answers or escalates to a human.

Full-stack Developer, Briki (insurtech startup) February–May 2025

React frontend, Node.js backend and PostgreSQL work; bug fixes and technical pitches.

Selected Software Projects

HeliOS: educational RISC-V kernel 2025–2026

Contributed to a collaborative C kernel with task scheduling, timer interrupts, a UART shell, synchronisation primitives and heap allocation; includes QEMU/OpenSBI smoke tests. Kernel, architecture and test scripts

ContratIA Abierta: public-procurement review tools 2026

Review pipeline for Colombian SECOP open data (Polars, PostgreSQL, MongoDB, FastAPI, Dash) with interpretable signals and traceable evidence for human review. Data pipeline and review dashboard